

$$1) \quad S = \sum_{i=1}^{\infty} b_i 2^{-i}$$

$$X(n, S) = b_n \quad n=1, 2, \dots$$

$$a) \quad S = b_1 2^{-1} + b_2 2^{-2} + \dots$$

$$= \frac{b_1}{2} + \frac{b_2}{4} + \dots$$

(not unique)

--- (1)

$$\text{Say } [1, 1, 0, 0, 0, \dots] \rightarrow S = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\text{Sample realization: } X(1, 3/4) = b_1 = 1$$

$$X(2, 3/4) = b_2 = 1$$

$$X(3, 3/4) = b_3 = 0$$

⋮

Same for rest.

$$b) \quad P(X(1, S) = 0) \quad \text{and} \quad P(X(1, S) = 0 \ \& \ X(2, S) = 0)$$

$$P(X(1, S) = 0) = P(0 \leq S < 1/2) = 1/2 \quad \text{--- (1)}$$

($\because$ half of sequences begin with 0)

$$P(X(1, S) = 0 \ \& \ X(2, S) = 0) = P(1/4 \leq S < 1/2) = 1/4 \quad \text{--- (1)}$$

($\because$ half of seq begin with 0, and half of half begin with 0)

$$2) a) I_n = [0 \ 0 \ 1 \ 0 \ 1 \ 1 \ \dots]$$

$$D_n = [-1 \ -1 \ 1 \ -1 \ 1 \ 1 \ \dots]$$

$$b) \text{ Mean: } E[D_n] = E[2I_n - 1] = 2E[I_n] - 1$$

I_n is a Bernoulli process, 1 occurs w.p. p .

$$\Rightarrow \text{Mean} = 2p - 1.$$

$$\text{Var}[D_n] = 2^2 \text{Var}[I_n] = 4p(1-p).$$

3. a) n even: $X_n = \pm 1$ w.p. $\frac{1}{2}$.

n odd: $X_n = \frac{1}{3}$ or -3 w.p. $\frac{9}{10}$ & $\frac{1}{10}$ resply.

$$\mu_x(t) = \frac{1}{2} \left(\frac{1}{2}(+1) + \frac{1}{2}(-1) \right) + \frac{1}{2} \left(\frac{9}{10} \cdot \frac{1}{3} + (-3) \cdot \frac{1}{10} \right) \\ = 0$$

Auto-correlation: $R_x(n_1, n_2) = E[X(n_1) X(n_2)]$ (since they are independent)

$$\Rightarrow R_x(n_1, n_2) = \begin{cases} 0 & \text{if } n_1 \neq n_2 \\ E[X^2(n_1)] = 1 & \text{for } n_1 = n_2 \end{cases}$$

$\Rightarrow$ Random process not strictly stationary since distribution changes at each point $(n_1, n_2, \dots)$

But R.P. WSS since mean is constant & auto-correlation is constant (weakly function of τ , most importantly not a function of $n_1/n_2/\dots$).

$$3. b. \quad X(t) = A + Bt + t^2$$

$$E(A) = E(B) = 0.$$

$$\Rightarrow \mu_X(t) = t^2 \quad \dots (0.5)$$

Auto correlation:

$$R_X(s, t) = E[(A + Bs + s^2)(A + Bt + t^2)] \quad \dots (0.5)$$

$$= E[A^2 + ABt + At^2 + ABs + B^2st + Bst^2 + As^2 + Bts^2 + s^2t^2]$$

$$= 1 + st + s^2t^2 \quad \dots (1)$$

$$\because E(A) = E(B) = 0 \Rightarrow E[A^2] = E[B^2] = 1.$$

$$\begin{aligned}
 \textcircled{5} \quad P &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left(A^2 \cos^2 \pi f_2 t + A^2 \cos^2 2\pi f_1 t \cos^2 2\pi f_2 t + \right. \\
 &\quad \left. 2 A^2 \cos^2 \pi f_2 t \cos 2\pi f_1 t \right) dt. \\
 &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left(\frac{A^2}{2} (1 + \cos 4\pi f_2 t) + \frac{A^2}{4} (1 + \cos 4\pi f_1 t) (1 + \cos 4\pi f_2 t) \right. \\
 &\quad \left. + \frac{2A^2}{2} (1 + \cos 4\pi f_2 t) \cos 2\pi f_1 t \right) dt
 \end{aligned}$$

$$= \frac{A^2}{2} + \frac{A^2}{4} = \frac{3A^2}{4}$$

Now Signal range : $[-m_p, m_p] = [2A, -2A]$

$$\Delta = \frac{2m_p}{L} = \frac{2 \cdot 2A}{2^n} = \frac{A}{2^{n-2}}$$

$$N_f = \frac{m_p^2}{3L^2} = \frac{(2A)^2}{3 \cdot 2^{2n}} = \frac{4A^2}{3 \cdot 2^{2n-2}}$$

$$\text{SNR} = \frac{\text{Signal power}}{\text{Noise power}}$$

$$= \frac{\frac{3A^2}{4}}{\frac{4A^2}{3 \cdot 2^{2n-2}}} = \frac{9}{16} 2^{2n-2}$$